

The contact-edge selector: why $\theta_c = \inf \mathcal{F}$ fails, and the closed-form tangency residual that replaces it

SpectralKM derivations

1 The symptom — retracted

An earlier version of this note opened by asserting that the model overpredicts the contact time by roughly a factor of two. That is retracted: it was an artefact of the comparison, not a property of the model.

Run at *identical* dimensionless parameters, this model and the reference implementation (harrislab-brown/BouncingDroplets, executed directly in MATLAB) agree:

metric	ours	reference	difference
contact duration (<code>primary_contact_time</code> / their $f > 0$)	3.6293	3.6530	0.65%
threshold, centre of mass below R	4.4432	4.5297	1.9%
$\min z_{cm}/R$	0.1670	0.1110	we penetrate less

at $We = 1.0959$, $Bo = 0.01661$, $Oh = 0.00617$ ($R = 0.35$ mm water, $U = 0.4759$ m/s).

Where the false symptom came from: the wrong physical problem. The dataset the comparison was built on is the **droplet-onto-solid-substrate** problem, not the droplet-onto-bath problem this model solves. The spreadsheet is named after the repository `LowWeberDropRebound`, whose `README` opens “MATLAB scripts for simulating the dynamics of liquid droplets impacting *solid substrates*”. Six independent checks agree:

Check	Result
that repo’s <code>sweeper_experiments.m</code>	hardcodes $Bo = 0.0189$ — exactly the single fixed Bo of the file’s <code>dns</code> and <code>km_model</code> blocks
<code>besselj</code> , <code>tanh</code> , <code>bath</code> in its solver	zero occurrences: no bath surface degrees of freedom at all, only the drop’s β_l
the bath paper’s Figure 5	water at $Bo = 0.017$, $Oh = 0.006$, with quasi-potential and DNS curves <i>only</i> — no “KM model” curve
the bath paper’s experiments	exactly <i>two</i> (Bo, Oh) points (water, oil); this file’s is a continuum, $Oh = 0.0139 \dots 0.7865$
the file’s <code>km_model</code> block	includes $Bo = 0$ and $Oh = 0$ rows, i.e. an inviscid zero-gravity sweep, not a two-fluid experiment
contact-time bands	disjoint (below)

The last check is the physical one, and it is decisive. On a *rigid wall* the rebound is set by the drop’s own $l = 2$ free oscillation, full period $2\pi/\sqrt{8} = 2.2214 t_\sigma$, with the classical superhydrophobic value $\approx 2.6 t_\sigma$ [3]; that file gives `km_model` = 2.41 at $We = 1$ and `dns` $\in [1.96, 3.36]$. A *bath* deforms and carries impact energy away as interfacial waves, so the drop rides down and back up and contact lasts markedly longer: the bath dataset (`lowWeberComparison.csv`, 153 runs at $N = 60$, 89 with a measured t_c) spans $[4.4408, 8.0946]$. The two bands do not overlap, so no amount of parameter matching reconciles them — they are answers to different questions. This is now pinned in `test/test_bath_reference.jl`.

At the bath reference’s own water point, $Bo = 0.016644$, $Oh = 0.006165$, it reports $t_c = 4.5286$; running the reference MATLAB directly gives 4.5297, and this model gives 4.4432 — squarely in the bath band. The earlier “puzzle” of 4.53 against a `km_model` value of 2.41, which appeared to contradict the paper’s own claim that t_c is nearly Oh -independent, was never a puzzle: those are a bath and a rigid wall.

Against experiment, not against another code. All of the above is model-against-model, which settles the problem identity and nothing else. The bath *experiments* are now in the repository as `bath_experiment_water.csv` and `bath_experiment_oil.csv`: 17 measured points with standard deviations over at least five trials each, extracted by `scripts/extract_bath_experiment.m` from the authors’ own `.fig` files. They were absent from the digitised CSV export that already existed, because the experimental points are stored as `errorbar` objects — one per point — and an export that walks only `line` children drops every one of them; that export holds the figure’s eight model curves and none of its five water points.

	We	t_c/t_σ	
experiment (water, $Bo = 0.017$, $Oh = 0.006$)	0.725	4.6628 ± 0.1887	≥ 5 trials
this model, <code>threshold_contact_time</code>	0.700	4.5039	
residual		−3.4%	0.84σ

Two things keep that number honest. The metrics are *not* the same: the experiment times the north pole across $z = 2R$, detachment not being optically resolvable, whereas the model convention — which `threshold_contact_time` implements — is the centre of mass across $z = R$, and the authors quote a typical 5% difference between the two for water. And the paper states that its own quasi-potential model and DNS “slightly underpredict the dimensionless contact time at intermediate We ”, attributing it to neglected nonlinear deformation. A residual of −3.4% is therefore consistent with the definition offset alone, and its sign is the one the authors report. Pinned in `test/test_bath_reference.jl`.

A second, compounding error. Independently of the wrong-problem error, the comparison script selected parameters by *binning on We alone*: it split $\log_{10} We$ into equal intervals over the 931 points and took the median We , Bo and Oh of whatever fell in each, so Bo and Oh were never matched to any actual measurement. A single window mixes fluids differing by 33–54 $\times$ in Oh , and the t_c values *within* one window already differ by 1.35–1.48 $\times$. That confound was noted in writing when the script was created and not acted upon for several rounds. It is fixed — `choose_points` now clusters on (We, Bo, Oh) jointly and reports each point’s Oh and Bo spread — but it was the smaller of the two mistakes.

What was eliminated along the way. Each by measurement, and each still worth knowing:

Candidate	Test	Outcome
metric definition	<code>threshold_contact_time</code> vs. <code>contact_time</code>	real, and it does differ
drop viscous closure	<code>:reid</code> vs. <code>:lamb</code>	4th digit
bath radius (reflected waves)	<code>debug_bath_radius_sensitivity.jl</code> $b = 6, 12, 20, 30$ with $M \propto b$	flat: 4.9402, 4.9613, 4.9614, 4.9614
contact-edge rule	<code>:crossing</code> (the reference's own rule)	unchanged to 3 digits

$b = 6$ remains unvalidated per `provenance.tex`; it is now measured as *not* affecting the contact time.

2 What is actually wrong

Tracing a contact (`debug_contact_exit_trace.jl`, `debug_feasibility_collapse.jl`):

- the net force $f \leq 0$ on **0 of 221** steps – it decays asymptotically to 1.4×10^{-8} from above and never reverses, so the $f \leq 0$ exit test never fires and contact terminates only at the θ_c floor;
- 63% of the contact duration occurs *after* the droplet has begun rising;
- at $t = 0.009$ the non-intersection predicate is false for all $\theta < 0.0492$ and $\inf \mathcal{F} = 0.0492$; one step later, at $t = 0.010$, it is true even at $\theta = 10^{-4}$, $\inf \mathcal{F}$ collapses by a factor of 200, and the patch never recovers.

None of this explains a contact-time error, because there is none to explain. What follows stands on its own: the rule is degenerate, that degeneracy is real and reproducible, and it produces a physically indefensible patch at low We . It is a defect worth fixing for its own sake, not the cause of a symptom that turned out to be an artefact.

The search is not at fault. A brute-force scan (`debug_selector_rate_limit.jl`) shows `select_theta_c` returning within 0.3–4% of the true $\inf \mathcal{F}$ at every step. The *rule* is at fault.

3 The degeneracy, stated precisely

Write the gap as $C(\theta, \tau) = \eta(r(\theta, \tau), \tau) - z_{cm} + z_d(\theta, \tau)$. The kinematic match imposes $C = 0$ on $[0, \theta_c]$; physical consistency requires $C < 0$ beyond the edge. Continuing just outside, $C(\theta) \simeq a(\theta - \theta_c)$ with

$$a(\theta_c) := \partial_\theta C|_{\theta_c}, \quad (1)$$

so non-overlap is the inequality $a(\theta_c) \leq 0$ and the feasible set is $\mathcal{F} = \{\theta_c : a(\theta_c) \leq 0\}$. The two rules are then

$$\text{current: } \theta_c = \inf \mathcal{F}, \quad \text{tangency (3): } a(\theta_c) = 0. \quad (2)$$

If $a(\theta_c) < 0$ for every $\theta_c > 0$ – the measured situation from $t = 0.010$ onward – then $\mathcal{F} = (0, \theta_{\max}]$ and $\inf \mathcal{F} = 0$, *regardless of the shape or magnitude of a .*

This is the whole failure. The infimum rule uses only the **sign** of a and never its magnitude, so it returns the same answer whether the surfaces are barely separated or widely so. It carries information only while the constraint is *binding*; the moment non-intersection goes inactive, it degenerates to $\theta_c = 0$ – not because the physics says the patch vanishes, but because the rule has nothing left to determine it with. Verified symbolically in `derivations/cas_tangency_residual.py`, which also exhibits an $a < 0$ example where $\inf \mathcal{F} = 0$ while $\arg \min |a|$ is interior.

Two further facts from the scan bear on the rule’s stated justification. Positivity ($p > 0$) holds *at* the collapsed patch, so complementarity – which `select_theta_c`’s docstring cites as its basis – does not forbid it. And the band’s upper edge is set by $f < 0$ and $p < 0$, i.e. by suction, not by `eq:check-monotone-r`. The rule’s original evidence was a single measurement at one step at onset, precisely where the constraint *is* active and the rule does work.

4 The closed-form tangency residual

`eq:tangency-selector` is $T = \partial_\theta C|_{\theta_c}$, currently evaluated in `src/residual.jl` by a centred finite difference with $h = 10^{-6}$.

A claim withdrawn. An earlier draft of this note asserted that differencing a Fourier–Bessel sum at $h = 10^{-6}$ “discards roughly half the available digits” and was therefore unfit to root-find. *That was wrong, and the test written to check it said so:* against a Richardson-extrapolated reference at $h = 10^{-3}$, the existing finite-difference residual agrees to `rtol` = 10^{-8} at $\theta = 0.25, 0.6, 1.1$ (`test/test_selector.jl`). Accuracy is not the problem. The closed form below is still worth having — it removes a step-size parameter, and it differentiates cleanly under `ForwardDiff` where a difference of a difference would not — but the case for it is structural, not numerical, and the fix to the selector does not depend on it. With $\xi(\theta) = 1 + \sum_l \beta_l P_l(\cos \theta)$, $r = \xi \sin \theta$, $z_d = -\xi \cos \theta$ and $\eta(r) = \sum_m a_m J_0(k_m r)$, the chain rule gives

$$T = \eta'(r) [\xi' \sin \theta + \xi \cos \theta] + [\xi \sin \theta - \xi' \cos \theta] \quad (3)$$

with

$$\xi'(\theta) = -\sin \theta \sum_l \beta_l P'_l(\cos \theta), \quad \eta'(r) = -\sum_m a_m k_m J_1(k_m r). \quad (4)$$

(3) is verified against symbolic differentiation of the composed expression (difference exactly zero), and against the existing numerical residual in `test/test_selector.jl`.

5 The pole degeneracy is physical, not algebraic

$T \equiv 0$ at $\theta = 0$ and $\theta = \pi$ for any state – `eq:tangency-degenerate`. Worth recording *how* the CAS established this, because the first attempt was wrong: with η left as an abstract function the CAS correctly **refused**, returning $T(0) = \xi(0) \eta'(0)$, since nothing in the chain rule makes $\eta'(0)$ vanish. It vanishes because of **axisymmetry**, through $J_1(0) = 0$; and ξ' carries a factor $\sin \theta$, so it vanishes at both poles too. Substituting the concrete expressions, $T \rightarrow 0$ at both poles for $l = 2$ and $l = 3$ and arbitrary $\beta_l, a_m, k_m, z_{cm}$.

Two implementation consequences follow. A seed is required at onset, which `onset_theta_c` already supplies from the geometric crossing $C = 0$; and the residual must be root-found *away* from $\theta = 0$, never bracketed from it.

6 The replacement rule

Select θ_c as a root of (3), retaining `eq:check-nonintersect` and `eq:check-monotone-r` as a *feasibility filter* over candidates – which is what `paper-formulation.tex` describes, and what [1] and [2] do. Because $\eta(r(\theta))$ is a sum of oscillatory J_0 terms, T can change sign more than once on $(0, \pi)$, so a plain bisection on a sign change is not safe; the `argmin` form of `eq:theta-c-argmin` over admissible candidates is used instead, seeded from the previous step and, at onset, from `onset_theta_c`.

7 The degeneracy is fatal at high Weber number: `:crossing` becomes the default

An earlier revision of this note closed by saying that the degeneracy, though real, changed no observable, and `DEFAULT_SELECTOR` was therefore left at `:feasible`. **That verdict was drawn from too narrow a sample and is retracted here.** It rested on two low-Weber water points, $We = 0.0231$ and $We = 0.9985$, where the two rules agree to three digits. They do. At high We they do not agree at all: one returns a physical trajectory and the other returns a droplet that sinks.

At $We = 7.307$ in 5 cSt silicone oil ($Bo = 0.0563$, $Oh = 0.0578$), `:feasible` holds contact for 21 steps — a duration of 0.194 against a measured 5.26 — then detaches, after which the droplet free-falls to $z_{cm} = -84.2$ by $\tau = 30$ and never re-contacts. No contact time exists to report. Under `:crossing` the same case gives contact over $(0, 4.229)$, a rebound to $z_{cm} = 2.239$, and further contacts at $\tau \in (18.33, 22.15)$ and $\tau > 29.38$: a physical multi-bounce trajectory. This is the degeneracy of §3 at full strength — once non-intersection stops binding, $\inf\{\theta : \text{feasible}\}$ is 0 regardless of the gap, and at high We that happens almost immediately rather than late in contact.

7.1 The change is a strict improvement, measured

The switch was gated on both observables the experiments report, at the default truncation ($M = 60$, $L = 120$, $N = 3$, $n_q = 200$), at five anchor points spanning both working fluids, with each rule run on identical hardware. Contact time is t_c/t_σ ; δ is the maximum penetration depth of the droplet’s south pole, taken over the *first* contact interval.

fluid	We	t_c :feasible	t_c :crossing	δ :feasible	δ :crossing
water	0.7251	4.49560	4.49657	0.66951	0.66900
water	1.9387	4.42119	4.42242	0.95812	0.95736
oil	1.2158	5.07868	5.08034	0.72344	0.72301
oil	3.9463	5.01559	5.01686	1.07481	1.07417
oil	7.3070	<i>none</i>	5.03505	0.30009	1.33536

Wherever `:feasible` produces an answer at all the two rules agree to four significant figures — at most 0.03% in t_c and 0.08% in δ — so `:crossing` cannot be said to regress any case that previously worked. Where `:feasible` fails it fails completely. That is a strict improvement, and it is why the default changes. The cost of the change to stored results is correspondingly small: any low- We number regenerated after this commit moves in its fourth digit, and any high- We oil number changes qualitatively because it was not physical before.

7.2 What the gate did *not* vindicate: δ

The same measurement exposes a separate deficit that is *not* selector-related, since it is identical to four figures under both rules. Against the published quasi-potential model (1PKM), interpolated

from the same figures the experimental points were read from:

fluid	We	δ here	δ 1PKM	δ measured
oil	1.2158	0.7230 (-1.60σ)	0.7415 (-1.30σ)	0.8229 ± 0.0626
oil	3.9463	1.0742 (-4.69σ)	1.2811 (-2.65σ)	1.5495 ± 0.1013
oil	7.3070	1.3354 (-4.40σ)	1.7042 (-2.15σ)	2.0558 ± 0.1637

Under-prediction of δ is a property of the reduced-model class, not of this package: over the twelve oil points 1PKM averages -1.64σ (-12.3%) and clears 2σ at only 8 of 12, while the DNS averages -0.71σ and clears it at all 12. But this package under-predicts δ roughly twice as far as 1PKM does at moderate and high We, while simultaneously predicting t_c *better* than 1PKM there (-0.85σ against -3.61σ at $We = 7.307$). A model that gets the contact duration right and the penetration depth wrong is saying something specific about where its linearisation fails, and that is open.

8 Scope

This note establishes that the current rule is degenerate, and §7 establishes that the degeneracy destroys the high-We solution outright, which is why `:crossing` is now the default. What remains open is the δ deficit above, which is selector-independent and roughly twice 1PKM's at moderate We. Any future claim here must be made against data for the *same physical problem* — a bath, not a solid substrate — and then *parameter-matched*, on Bo and Oh and not We alone. Nor does this note revisit the upper edge of the admissible band, where $f < 0$ and $p < 0$ already exclude wide patches; whether that exclusion is itself too aggressive is a separate question, and the δ deficit makes it a more interesting one — too aggressive an upper edge would under-predict penetration.

References

- [1] L. F. L. Alventosa, R. Cimpeanu, D. M. Harris, *Inertio-capillary rebound of a droplet impacting a fluid bath*, JFM (2023).
- [2] C. A. Galeano-Ríos, P. A. Milewski, J.-M. Vanden-Broeck, *Non-wetting impact of a sphere onto a bath*, JFM (2017).
- [3] D. Richard, C. Clanet, D. Quéré, *Contact time of a bouncing drop*, Nature **417**, 811 (2002).